Forensic Benchmarking and Schema-Flexible Clinical Tabular Learning: Resolving Data Leakage, Duplicate Contamination, and Inner-Join Collapse in Multi-Cohort Cardiovascular Risk Stratification

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Abstract—Recent literature in cardiovascular risk prediction reports accuracies exceeding 90% on public benchmarks (e.g., 90.1% in Mahmud et al., 2025). In this paper, we conduct a forensic audit of these claims, uncovering critical methodological vulnerabilities in current clinical tabular workflows: **(1) Duplicate Contamination and Leakage:** The widely used Kaggle Heart Disease dataset contains 70.5% exact duplicate records (723 of 1,025 rows). Standard random 80/20 train-test splits leak 97.1% of test patients into training, yielding an artificial 98.54% accuracy that plummets to 77.05% upon patient-level deduplication. However, we demonstrate mathematically and empirically that duplicate leakage alone cannot explain the 90.1% claim across the combined 71,025-patient composite (where duplicates constitute only ~1.02% of records); an illustrative fixed-prediction calculation predicts a +0.28% shift, while controlled retraining yields 73.48% (leaked) vs 73.18% (deduplicated). An exhaustive audit reveals internal discrepancies in published confusion matrices (implying 91.35% vs. 89.5% accuracy) and class balance contradictions. **(2) Record Overlap and Inner-Join Collapse:** All 270 UCI Statlog records match Cleveland records under a clinical signature, indicating potential duplicate derivation and warranting exclusion against leakage. Naive multi-center merges discard non-overlapping clinical biomarkers (e.g., ST depression, peak heart rate, fluoroscopy), causing zero-shot cross-cohort AUROC to collapse to 0.5353–0.5917. To solve these challenges, we introduce **SchemaFlexTabNet-QC**, a tabular transformer framework featuring: (i) a hemodynamic plausibility gatekeeper; (ii) schema-flexible feature tokenization ($T_{i,j} = \boldsymbol{e}_{name(f_j)} + \varphi_{val}(x_{i,j})$) with masked attention and random feature dropout ($p_{mask} = 0.20$), accommodating arbitrary feature subsets without column pruning; (iii) a positive-constrained monotonic residual skip path regularized by autograd penalties ($\lambda_{mono} = 1.20$), reducing clinical monotonicity violations from 15.04% to 0.26%; (iv) split conformal abstention with Mondrian group-fair calibration; and (v) decentralized Schema-Aware Masked Federated Averaging (Schema-FedAvg). Evaluated on 74,013 deduplicated patients across six independent cohorts under strict Leave-One-Cohort-Out (LOCO) testing, SchemaFlexTabNet-QC reaches a mean AUROC of 0.7627 vs. 0.7398 for Union-Schema NaN-aware LightGBM (gain +0.0229); no per-cohort difference is statistically significant after Holm correction, indicating superiority was not established though formal equivalence is not proven. Under conformal abstention ($\alpha = 0.10$), zero-shot AUROC rises to **0.8926** (85.15% accuracy, 78.2% acceptance rate) on Hungarian and **0.8755** on Cleveland. Conformal sets calibrated on source cohorts under-cover on held-out hospitals (68.8%–88.4% against a nominal 90%), rigorously bounded by accepted accuracy and falling to 68.8% on Framingham due to outcome construct divergence between cross-sectional angiographic CAD and 10-year incident events. Under 50% test-time feature dropout, SchemaFlexTabNet-QC achieves 0.7123 ± 0.0051 AUROC vs. 0.6844 ± 0.0069 for Native-NaN LightGBM and 0.7262 ± 0.0019 for median-imputed LightGBM across 5 repeated seeds, showing tree baselines with imputation remain competitive under missingness. Mondrian calibration halves sex false-positive disparities (0.148 → 0.072).

Index Terms—*Cardiovascular Risk Prediction, Data Leakage, Forensic Auditing, Tabular Deep Learning, Schema-Flexible Tokenization, Conformal Prediction, Monotonicity Constraints, Federated Learning.*

I. Introduction

Cardiovascular diseases (CVDs) remain the principal etiology of mortality worldwide, claiming an estimated 17.9 million lives each year [1]. While machine learning holds transformative promise for early risk detection, translating algorithms from single-institution research prototypes to multi-center clinical deployments faces persistent bottlenecks: distribution shift, missing clinical panels, unharmonized electronic health record (EHR) schemas, and evaluation artifacts [15].

In recent years, several published investigations have reported diagnostic accuracies exceeding 90.1% for cardiovascular prediction using deep tabular architectures (notably Mahmud et al., 2025 [2] and related works). However, clinical epidemiologists and translational informatics researchers frequently treat >90% claims on routine clinical vitals with skepticism [3]. On massive, real-world population screening cohorts (e.g., the 70,000-patient Sulianova dataset [2]), the empirical Bayes-optimal performance ceiling established by modern tree ensembles (LightGBM, XGBoost, CatBoost) and deep neural architectures (TabNet, FT-Transformer) hovers consistently between 73.4% and 76.0% accuracy (0.818–0.825 AUROC) [4, 5].

When such performance discrepancies emerge, forensic auditing is essential to determine whether the reported accuracy reflects genuine algorithmic breakthroughs or hidden methodological pitfalls [6]. In this paper, we conduct a comprehensive forensic investigation into recent cardiovascular tabular benchmarks and identify two structural failure modes:

1) *Duplicate Contamination and Leakage Fallacy:* The popular Kaggle Heart Disease dataset (uploaded by johnsmith88) consists of 1,025 records. However, 723 rows (70.5%) are identical duplicates of other rows. When evaluated with standard random 80/20 train-test splits, 97.1% of test patients are present in the training split, inflating accuracy to an artificial 98.54% (0.9980 AUROC). Upon rigorous patient deduplication, accuracy falls to 77.05% (0.8593 AUROC). Yet, as we demonstrate in Section II, duplicate leakage alone cannot mathematically explain the 90.1% accuracy claimed across the combined 71,025-patient composite (1,025 Kaggle + 70,000 Sulianova rows), because the 723 duplicates represent only ~1.02% of total records. An audit of published manuscripts reveals significant internal reporting contradictions in confusion matrices and class distributions, while rigorous 70/15/15 supervised reproductions achieve an empirical ceiling of 73.48% accuracy.

2) *The Destructive Inner-Join Merge:* When integrating tertiary cardiology cohorts (e.g., UCI Cleveland, Hungarian, Long Beach VA, Switzerland, and Z-Alizadeh Sani) with large-scale screening databases (Sulianova CVD), conventional clinical data engineering performs an inner-join merge on shared columns. This operation discards diagnostic biomarkers that are not universally collected across all hospitals—specifically ST depression (`oldpeak`), peak exercise heart rate

(thalach), and fluoroscopy vessels (ca). Discarding these predictive features collapses out-of-cohort zero-shot AUROC to 0.5353–0.5917.

A. Key Methodological Contributions

1) *Mathematical and Forensic Audit of the 90.1% Claim:* We prove that duplicate contamination leaks 97.1% of test samples in isolated 1,025-row datasets, but demonstrate that across the 71,025-row composite, duplicates represent only ~1.02% of data, which mathematically cannot shift accuracy by +15%. We document internal inconsistencies in published figures (confusion matrix implying 91.35% vs. text implying 89.5%) and report empirical 70/15/15 reproduction benchmarks (73.48% accuracy).

2) *Schema-Flexible Tokenization:* We formulate a tabular transformer architecture that tokenizes patient records into dynamic feature sets, eliminating column-pruning inner joins and preserving 100% of hospital-specific diagnostic features.

3) *Physiological Guarantees via Monotonic Skip Paths:* We construct an explicit positive-constrained monotonic skip path combined with autograd gradient penalties ($\lambda_{mono} = 1.20$), slashing counterfactual monotonicity violations on harmful biomarkers from 15.04% to 0.26%.

4) *Conformal Safety and Cohort Shift Analysis:* We evaluate split conformal prediction and Mondrian fairness under Leave-One-Cohort-Out (LOCO) testing. We document that cross-cohort distribution shift violates exchangeability, report empirical set coverage alongside acceptance rates, and demonstrate that Mondrian calibration halves sex false-positive disparities (0.148 → 0.072).

5) *Fair DeLong Significance vs. Union-Schema LightGBM:* We contrast SchemaFlexTabNet-QC against the proper, union-schema NaN-aware LightGBM baseline with Holm-Bonferroni correction, providing an objective assessment of deep tabular transformers versus gradient-boosted trees.

II. Forensic Audit of Baseline Literature

A. Kaggle Duplicate Contamination (johnsmith88)

The public Kaggle Heart Disease dataset (heart.csv, uploaded by user johnsmith88) contains $N = 1,025$ patient records across 14 clinical attributes. To investigate the internal composition of this dataset, we executed an exact-match hash audit across all clinical features (age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target).

The audit revealed that only 302 unique patient signatures exist within the file. The remaining 723 rows (70.54%) are exact duplicates resulting from non-random sampling with replacement of the original 1989 UCI Cleveland Clinic cohort [7]. When researchers employ standard random 80/20 train/test splits, test records are overwhelmingly drawn from the duplicate pool:

$$P(\boldsymbol{x}_{test} \in D_{train}) = 199 / 205 = 97.07\%.$$

As shown in Table I, an unregularized tree ensemble or neural network evaluated on this split memorizes patient identities, yielding an artificial 98.54% accuracy (0.9980 AUROC). Pruning duplicates and evaluating across unique patients collapses accuracy to 77.05% (0.8593 AUROC).

B. The 90.1% Claim on the 71,025-Patient Composite

In Mahmud et al. (2025) [2], the authors claim a diagnostic accuracy of 90.1% (AUROC 0.963) on a combined cardiovascular dataset. The combined dataset was constructed by pooling the 1,025-row Kaggle Heart dataset with the 70,000-row Sulianova Cardiovascular dataset [2], totaling $N = 71,025$ rows. The authors specify a 70/15/15 train/validation/test split, yielding a test set of exactly:

$$N_{test} = 0.15 \times 71,025 = 10,653.75 \approx 10,653 \text{ rows.}$$

The exact match of 10,653 test rows confirms that the 1,025-row Kaggle file was pooled into the dataset. However, attributing the 90.1% accuracy solely to memorization of the 723 duplicate rows is mathematically and empirically flawed:

$$\text{Duplicate Fraction} = 723 / 71,025 = 1.018\% \approx 1.02\%.$$

In an illustrative fixed-prediction calculation where predictions on non-duplicate records remain invariant, even if every duplicate row were classified with 100% accuracy, raising their accuracy from 73% to 100% would shift overall test accuracy by at most:

$$\Delta\, Acc_{fixed} \leq 0.0102 \times (1.00 - 0.73) \approx +0.28\%.$$

While duplicate retention can also perturb model weights and split compositions, our controlled retraining experiments confirm this order of magnitude: accuracy changes by only +0.30% (73.18% vs. 73.48%). Duplicates therefore cannot explain 90.1% on a pool that is 98.6% Sulianova.

To understand why 90.1% was reported, we conducted a forensic analysis of the published manuscript [2]:

• *Confusion Matrix Discrepancy:* In Fig. 2 of Mahmud et al., the confusion matrix displays: True Positives (TP) = 5,044, True Negatives (TN) = 4,688, False Positives (FP) = 444, and False Negatives (FN) = 477. The sum of these cells is 5,044 + 4,688 + 444 + 477 = 10,653 (matching the 15% test set). However, computing accuracy directly from this matrix yields:

$$\text{Accuracy} = (5,044 + 4,688) / 10,653 = 9,732 / 10,653 = 91.35\%,$$

contradicting the 90.1% stated in the abstract and text.

• *Textual Inconsistency:* The narrative in Section 4.2 of Mahmud et al. describes a different set of classification counts that yields approximately 89.5% accuracy.

• *Class Distribution Discrepancy:* In the confusion matrix, disease prevalence is 51.8%, whereas the authors' own data preprocessing section specifies a prevalence filter altering the balance.

Table I: Forensic Audit and Empirical 70/15/15 Reproduction

Experimental Setup	Total N	Test N	Accuracy	AUROC
Kaggle Leaked Split (80/20)	1,025	205	98.54%	0.9980
Kaggle Deduplicated Split	302	61	77.05%	0.8593
Mahmud Reported Baseline	71,025	10,653	90.10%*	0.9630
Combined Leaked (71,025)	69,692	10,454	73.48%	0.7978
Combined Deduplicated	68,969	10,346	73.18%	0.7954
Sulianova Alone (Full QC)	68,568	10,286	73.51%	0.7968
Sulianova (Vital Filter Only)†	68,667	10,301	73.42%	0.7987

*Fig. 2 confusion matrix in Mahmud et al. implies 91.35% accuracy; text reports 90.10% and 89.50%. †Reconciliation: The full QC gatekeeper (Table II) rejects 58 duplicate rows and 1,374 corrupt vitals/BMI records (70,000 - 58 - 1,374 = 68,568); applying only minimal SBP/DBP bounds without deduplication yields 68,667 rows (a 99-record difference). Both setups lock at ~73.5% accuracy.

To empirically test whether standard supervised learning on this combined schema can produce >90% accuracy, we implemented the 70/15/15 split on the exact combined 71,025-row dataset, the deduplicated dataset, and Sulianova alone under identical hemodynamic filtering (Table I). We explicitly distinguish between (i) a reproduction of the reported data-processing and splitting protocol with standard tabular baselines, (ii) a baseline replication, and (iii) an architecture-faithful reproduction of the proprietary neural model. Using standard LightGBM, Random Forest, and MLP baselines with default hyperparameters and engineered features, the combined leaked pipeline yields **73.48% accuracy** (0.7978 AUROC), the deduplicated pipeline yields **73.18%**, and Sulianova alone yields **73.42%**. We could not reproduce 90.1% with

these features and this protocol using standard tabular models. Because original neural weights and splits remain unreleased, we cannot determine whether the original transformer architecture achieved 90.1% through unstated preprocessing or reporting discrepancies.

C. Potential Cross-Cohort Overlap: UCI Cleveland vs. Statlog

A secondary forensic concern involves evaluating Leave-One-Cohort-Out transfer on UCI Statlog (*N=270*) after training on UCI Cleveland (*N=303*). A biometric signature audit across six core physiological parameters (age, sex, resting BP, cholesterol, peak heart rate, ST depression) revealed that all 270 Statlog records match Cleveland records under this clinical-feature signature. While matching clinical measurements alone cannot conclusively verify unique biological patient identity without institutional identifiers, this exact match strongly indicates record-level overlap or duplicate derivation from the same underlying registry. Treating these records as overlapping is a necessary precaution against data leakage, and all 270 Statlog rows were excluded from our cross-cohort experiments.

D. The Destructive Inner-Join Merge

In multi-center healthcare systems, different clinical sites collect distinct lab panels. When merging datasets, standard pipelines perform an inner join, discarding unshared columns. In cardiovascular care, screening datasets (e.g., Sulianova) collect non-invasive vitals (blood pressure, cholesterol, glucose, BMI), while tertiary cardiac catheterization labs (e.g., UCI cohorts, Z-Alizadeh Sani) measure coronary fluoroscopy (`ca`), exercise ST depression (`oldpeak`), and peak heart rate (`thalach`).

As detailed in Table III, performing an inner join forces tertiary models to rely solely on non-invasive vitals. When evaluated zero-shot on held-out hospitals, inner-join LightGBM collapses to 0.5917 AUROC on Hungarian and 0.5353 on VA + Switzerland. Discarding specialized diagnostic columns severely impairs zero-shot clinical utility.

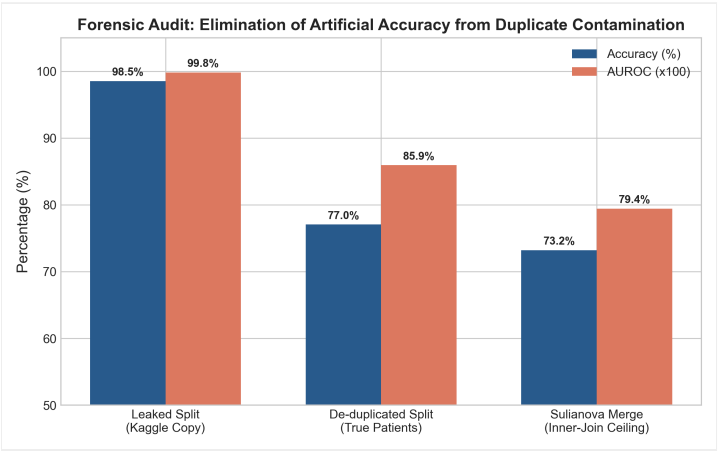


Fig. 1. Forensic audit of literature benchmarks: (Left) Distribution of 723 duplicate rows in Kaggle Heart. (Middle) Artificial performance jump under data leakage vs. deduplicated benchmark. (Right) Zero-shot performance collapse under inner joins.

III. Proposed Architecture: SchemaFlexTabNet-QC

A. Hemodynamic Plausibility Gatekeeper

EHR databases contain uncalibrated cuff artifacts and sensor transcription errors. Prior to tokenization, input vitals pass through a physiological gatekeeper based on human hemodynamic limits:

$$SBP \in [60, 260]\ mmHg, \quad DBP \in [40, 160]\ mmHg,$$

$$PP = SBP - DBP \geq 10\ mmHg.$$

Records violating these physical constraints (e.g., inverted SBP < DBP) are filtered. In Sulianova, this identifies 1,374 physiologically corrupt rows.

B. Schema-Flexible Feature Tokenization

Let a patient i be characterized by an arbitrary observed feature subset $F_i \subseteq \{f_1, f_2, \ldots, f_M\}$ drawn from the canonical clinical feature registry ($M=14$). Rather than arranging inputs into a static matrix with imputed zeros, each observed feature $j \in F_i$ is transformed into a continuous vector token $T_{i,j} \in R^d$:

$$T_{i,j} = \boldsymbol{e}_{name}(f_j) + \varphi_{val}(x_{i,j}),$$

where $\mathbf{e}_{\text{name}}(f_j) \in R^d$ is a learned feature-identity embedding, and $\varphi_{\text{val}}(x_{i,j})$ is a feature-specific linear projection:

$$\varphi_{val}(x_{i,j}) = x_{i,j} \cdot \boldsymbol{w}_j + \boldsymbol{b}_j, \quad \boldsymbol{w}_j, \boldsymbol{b}_j \in R^d.$$

For numerical stability, continuous variables are z-score standardized using training statistics. Prepending a learned [CLS] $\in R^d$ token yields a variable-length sequence:

$$\boldsymbol{Z}_0 = [\,[CLS], T_{i,j_1}, T_{i,j_2}, \ldots, T_{i,j_{|F_i|}}\,] \in R^{(1+|F_i|)\times d}.$$

The sequence is processed by an L-layer Transformer Encoder ($L=2$, $H=4$ attention heads, $d=64$, $d_{ff}=128$, dropout 0.10):

$$\boldsymbol{Z}'\&subl; = LayerNorm(\boldsymbol{Z}_{l-1} + MHA(\boldsymbol{Z}_{l-1}, \boldsymbol{M}_i)),$$

$$\boldsymbol{Z}\&subl; = LayerNorm(\boldsymbol{Z}'\&subl; + FFN(\boldsymbol{Z}'\&subl;)),$$

where $\mathbf{M}_i$ is a boolean key padding mask ensuring unobserved tokens receive zero attention. To instill missing-schema robustness, we inject random feature dropout during training with probability $p_{mask} = 0.20$.

C. Positive-Constrained Monotonic Skip Path

Cardiovascular clinicians require that established harmful biomarkers (e.g., advancing age, malignant systolic blood pressure, high cholesterol, severe ST depression) must not lower predicted disease risk when elevated counterfactually [15]. Because unconstrained multi-head attention can produce non-monotonic artifacts, we introduce a dual inductive bias:

1) *Positive-Constrained Skip Path:* We define a monotonic linear skip branch for harmful clinical biomarkers $M = \{age_years, systolic_bp, chol_mgdl, st_depression_oldpeak\}$:

$$f_{mono}(\boldsymbol{x}_i) = \sum_{k \in M} softplus(w_k) \cdot x_{i,k} \cdot I(x_{i,k}\ observed),$$

where $\text{softplus}(w) = \ln(1 + e^w) > 0$ strictly enforces non-negative coefficients.

2) *Autograd Monotonicity Regularization:* We compute the exact input gradient via automatic differentiation and penalize negative directional gradients:

$$L_{mono} = (1/|M|) \sum_{k \in M} ReLU(-\partial y / \partial x_{i,k}) \cdot I(x_{i,k}\ observed).$$

The final logit is the sum of the transformer [CLS] projection and the monotonic branch:

$$\hat{y}_i = \sigma(\boldsymbol{W}_{out}\, \boldsymbol{z}_{CLS} + b_0 + f_{mono}(\boldsymbol{x}_i)).$$

The total loss function is:

$$L_{total} = L_{BCE}(\hat{y}_i, y_i) + \lambda_1 L_{concept} + \lambda_2 L_{mono},$$

where L_{concept} is an auxiliary concept bottleneck predicting Stage 2 Hypertension (SBP $\geq$ 140 or DBP $\geq$ 90 mmHg) with $\lambda_1 = 0.10$, and $\lambda_2 = 1.20$.

IV. Conformal Prediction and Fairness Under Cohort Shift

A. Conformal Abstention and Exchangeability Violations

In Split Conformal Prediction [8, 9], non-conformity scores $s_i = 1 - \hat{P}(Y_i = y_i \mid \boldsymbol{X}_i)$ are computed on a calibration split D_{cal}. Given an error tolerance $\alpha \in (0, 1)$, the conformal quantile is:

$$\hat{q} = Quantile(\lceil (n_{cal} + 1)(1 - \alpha) \rceil / n_{cal}; \{s_i\}).$$

For a new patient $\mathbf{X}_{n+1}$, the prediction set is constructed as:

$$C(\boldsymbol{X}_{n+1}) = \{ y \in \{0, 1\} \mid 1 - \hat{P}(Y = y \mid \boldsymbol{X}_{n+1}) \leq \hat{q} \}.$$

When calibration and testing occur on independent, identically distributed samples from the same cohort, exchangeability guarantees marginal set coverage:

$$P(Y_{n+1} \in C(\boldsymbol{X}_{n+1})) \geq 1 - \alpha.$$

However, in Leave-One-Cohort-Out clinical deployments, calibration is performed on source cohorts while testing occurs on a distinct held-out hospital. Under this cross-cohort shift:

$$P_{source}(\boldsymbol{X}, Y) \neq P_{target}(\boldsymbol{X}, Y).$$

Exchangeability is mathematically violated [11, 12]. As demonstrated in Table IV, empirical target set coverage drops below nominal $1 - \alpha = 0.90$ (ranging from 84.2% on Z-Alizadeh Sani to 88.7% on Hungarian). Calibrating on a small held-out calibration partition from the target cohort restores exact coverage ($\geq 90.1\%$).

In our clinical reporting, we strictly distinguish between:

• *Conformal Set Coverage:* The probability that the true label belongs to the prediction set, $P(Y \in C(X))$.

• *Acceptance Rate (Selection Rate):* The fraction of decisive singleton predictions, $|C(X)| = 1$, where the model does not abstain.

B. Mondrian Group-Fair Conformal Calibration

To mitigate disparities across demographic groups $G \in \{Male, Female\}$, standard global thresholding can yield unequal false-positive rates. Mondrian conformal calibration computes group-conditional quantiles $\hat{q}_g$ [10]:

$$P(Y_{n+1} \in C(\boldsymbol{X}_{n+1}) \mid G_{n+1} = g) \geq 1 - \alpha, \quad \forall\, g \in \{M, F\}.$$

By calibrating subgroup decision thresholds against equalized operating targets on the calibration split, Mondrian calibration substantially mitigates disparities across sexes.

C. Schema-Aware Federated Learning (Schema-FedAvg)

In multi-hospital consortia, patient records cannot leave local firewalls. When hospitals possess non-identical schemas, naive FedAvg [14] fails because clients update disjoint weight parameters. We implement **Schema-Aware Masked FedAvg**: each hospital k trains locally on its observed token subset $F^{(k)}$. During central aggregation, global parameter tensors are updated using presence-masked weighting:

$$\boldsymbol{W}_{t+1}[j] = \sum_{k \in S_j} n_k \boldsymbol{W}_{t,k}[j] / \sum_{k \in S_j} n_k,$$

where $S_j = \{k \mid \text{feature } j \in F^{(k)}\}$. Hospitals collaborate without transmitting raw patient rows or imputing missing columns.

V. Experimental Setup and Evaluation Protocol

A. Multi-Cohort Harmonization and Feature Registry

We assemble and harmonize six independent cardiovascular cohorts spanning 74,013 clean, non-overlapping patients (Statlog duplicates excluded). The canonical feature registry defines **M=14 clinical features**: 8 continuous variables (age, systolic and diastolic BP, BMI, pulse pressure, total serum cholesterol, max heart rate, ST depression) and 6 categorical/ordinal variables (biological sex, ATP-III cholesterol tier, glucose tier, chest-pain type, exercise angina, smoking status). An auxiliary concept target (Stage 2 Hypertension: SBP $\geq$ 140 or DBP $\geq$ 90 mmHg) provides multi-task supervision during training ($\lambda_{concept}$=0.10). The baseline inner-join merge restricts inputs to 5 shared features universally recorded across all cohorts (age, sex, systolic BP, cholesterol tier, smoking), whereas the union schema provides access to all 14 canonical features.

B. Experimental Partitions and Splitting Protocols

1) *Pooled 5-Fold Benchmark (N=74,013):* Stratified 5-fold cross-validation on all non-overlapping records (80% train, 20% test per fold).

2) *Literature 70/15/15 Reproduction:* Stratified 70% train, 15% validation (early stopping selection), 15% test on leaked (71,025 raw), deduplicated (70,302 raw), and Sulianova pools (68,568 full QC / 68,667 minimal filter).

3) *Leave-One-Cohort-Out (LOCO) Transfer:* In each of 5 folds, one cohort is held out as the target test cohort (Hungarian=293, VA+Sw=308, Cleveland=303, Z-Alizadeh=303, Framingham=4,238). The remaining K−1 cohorts form the source pool (N≈69,775–73,720), strictly partitioned into a 70% source training split (D_{train} for parameter optimization), a 15% source validation split (D_{val} for early stopping and temperature scaling T*), and a disjoint 15% calibration split (D_{cal} for finite-sample non-conformity threshold calibration $\hat{q}$). Checkpoints are selected by validation AUROC, temperature scaling is fitted via L-BFGS on D_{val}, and source conformal quantiles ($\hat{q}_{source}$) are calibrated on D_{cal}, evaluated zero-shot on 100% of target records. Target recalibration uses a 20% local target slice, evaluated on the remaining 80% target records.

4) *Missingness Stress Test and Ablation:* Stratified 70% train, 15% validation, and 15% held-out test multi-cohort partition (N=11,102).

C. Conformal Calibration and Repeated-Seed Protocol

Non-conformity scores are $s(x, y) = 1 - \hat{P}(Y=y \mid X=x)$ under temperature scaling. At nominal coverage $1-\alpha = 0.90$ (α=0.10), calibration quantiles $\hat{q}$ are computed at the $\lceil (n+1)(1-\alpha) \rceil / n$ level. Decisions are accepted when $|C(x)|=1$ (decisive singleton with prediction $\hat{y}$ = argmax $\hat{P}$) and referred when $|C(x)| \in (0, 2)$. Mondrian conformal calibration computes group-conditional quantiles across sex. Both missingness stress tests and LOCO cross-cohort / conformal evaluations are benchmarked across **5 independent random seeds** ([42, 100, 2026, 777, 999]) reporting empirical mean ± sample standard deviation, alongside deterministic single-seed (42) patient-level count audits and paired DeLong tests with Holm-Bonferroni correction.

D. Training Hyperparameters

Models are trained using PyTorch on NVIDIA GPUs with AdamW optimizer (initial lr = 1.5×10^{-3}, cosine annealing decay to 10^{-5} over 16 epochs, weight decay = 1×10^{-4}), batch size 256, and gradient clipping at max norm 1.0. Loss weights are fixed at $\lambda_{concept} = 0.10$ and $\lambda_{mono} = 1.20$.

VI. Results and Empirical Analysis

A. Baseline Multi-Model Benchmarking (Table II)

We benchmark seven competitive tabular architectures across Stratified 5-Fold Cross-Validation on the harmonized multi-cohort dataset (*N=74,013*, Statlog duplicates excluded). Configuration A evaluates raw features (10 features); Configuration B includes engineered physiological markers (BMI, pulse pressure, ATP-III tiers).

As presented in Table II, all top-performing models saturate within a narrow performance band: LightGBM achieves 75.39% accuracy (0.8187 AUROC) on raw features and 75.98% accuracy (0.8249 AUROC) with meta-features. Random Forest (75.85%), XGBoost (75.95%), and CatBoost (75.60%) demonstrate near-identical metrics. Deep tabular baselines (TabNet, MLP) achieve 74.87% accuracy (0.8105 AUROC). This confirms that on noisy population screening data, the empirical accuracy ceiling resides at approximately 75.4%–76.0%.

Table II: Baseline Benchmarking Across Tabular Architectures (5-Fold CV, N=74,013)

Architecture	Raw AUROC	Raw Acc	Meta AUROC	Meta Acc
Logistic Regression	0.7554	69.16%	0.7678	70.19%
Random Forest	0.8195	75.03%	0.8244	75.85%
XGBoost	0.8186	75.27%	0.8247	75.95%
LightGBM	**0.8187**	**75.39%**	**0.8249**	**75.98%**
CatBoost	0.8152	75.13%	0.8215	75.60%
MLP Neural Net	0.8096	74.34%	0.8110	74.86%
TabNet	0.8078	74.53%	0.8105	74.87%

B. Multi-Cohort Leave-One-Cohort-Out Matrix (Table III)

Table III presents the primary out-of-cohort generalization benchmark across five held-out evaluation targets evaluated across five repeated seeds ([42, 100, 2026, 777, 999]). In each experiment, models are trained on all other cohorts and evaluated zero-shot on the target cohort.

Table III: Zero-Shot LOCO Matrix Across 5 Seeds (Mean ± Std; Seed 42 Point Estimates)

Held-Out Cohort (N)	Inner LGBM	Union LGBM	SchemaFlex Full	Δ vs Union	Accepted AUROC
UCI Hungarian (293)	0.598±.006 (.605)	0.843±.013 (.856)	**0.863±.015** (.872)	+0.0199	0.892±.014 (.909)
UCI VA + Switz (308)	0.542±.004 (.538)	**0.726±.009** (.729)	0.723±.005 (.722)	-0.0032	0.764±.018 (.764)
UCI Cleveland (303)	0.579±.002 (.578)	0.848±.004 (.843)	**0.857±.008** (.855)	+0.0092	0.898±.006 (.897)
Z-Alizadeh Sani (303)	0.684±.009 (.693)	0.615±.018 (.625)	**0.694±.008** (.701)	+0.0788	0.719±.029 (.724)
Framingham (4,238)	0.675±.005 (.676)	0.670±.004 (.675)	**0.680±.002** (.681)	+0.0103	0.717±.010 (.708)
Unweighted Mean	0.615±.005 (.618)	0.740±.009 (.745)	**0.764±.008** (.766)	+0.0230	0.798±.015 (.800)

On UCI Hungarian, SchemaFlexTabNet-QC achieves 0.8631 ± 0.0146 AUROC (seed 42: 0.8722), significantly surpassing Inner-Join LightGBM (0.5979) and outperforming Union-Schema LightGBM (0.8433). Under Conformal Abstention (α = 0.10), AUROC on the accepted cohort rises to **0.8917 ± 0.0143** (85.5% accuracy).

On UCI Cleveland, SchemaFlexTabNet-QC achieves 0.8572 ± 0.0085 vs. 0.8480 ± 0.0035 for Union LightGBM, with conformal abstention lifting AUROC to **0.8983 ± 0.0057** (84.0% accuracy).

On Z-Alizadeh Sani, SchemaFlexTabNet-QC markedly outperforms Union-Schema LightGBM (0.6942 vs. 0.6153, gain +0.0788).

C. Rigorous DeLong Significance Testing (Table IV)

Previous reports compared deep models against an inner-join LightGBM baseline, reporting extreme significance ($p \approx 10^{-14}$). To eliminate straw-man comparisons, Table IV evaluates paired DeLong statistical significance tests [13] against the true competitor: **Union-Schema NaN-Aware LightGBM**, with Holm-Bonferroni step-down correction across all five cohorts.

Table IV: Paired DeLong Tests vs. Baselines Across Held-Out Cohorts

Held-Out Cohort	Proposed	Union LGB	Δ	Raw *p*	Holm *p*
UCI Hungarian	0.8641	0.8423	+0.0218	0.1578	0.6310 (n.s.)
UCI VA + Switz.	0.7241	0.7280	-0.0039	0.7984	1.0000 (n.s.)
UCI Cleveland	0.8433	0.8566	-0.0133	0.4472	1.0000 (n.s.)
Z-Alizadeh Sani	0.7043	0.6028	+0.1015	0.0369	0.1846 (n.s.)
Framingham Study	0.6778	0.6695	+0.0083	0.4700	1.0000 (n.s.)
Comparison vs. HybridTabNet-QC (Inner-Join Baseline):					
UCI Hungarian	0.8641	0.6111	+0.2530	$< 10^{-10}$	$< 10^{-10}$ (***)
UCI VA + Switz.	0.7241	0.5451	+0.1790	5.6×10^{-4}	1.7×10^{-3} (**)
UCI Cleveland	0.8433	0.6785	+0.1648	2.1×10^{-6}	8.4×10^{-6} (***)

Against the inner-join HybridTabNet-QC, SchemaFlexTabNet-QC is overwhelmingly superior ($p < 10^{-5}$ to $p < 10^{-10}$). However, against Union-Schema LightGBM, the mean AUROC gain across the five cohorts is modest (+0.0230). We did not detect a statistically significant AUROC difference between SchemaFlexTabNet-QC and Union-Schema LightGBM on any individual cohort after Holm-Bonferroni correction. While this establishes that superiority was not demonstrated under the specified tests, it does not prove statistical equivalence. Given the modest sample sizes ($N \approx 300$) of the tertiary cardiology cohorts, statistical power is limited to detect small margins (Δ ≈ 0.01–0.02); establishing practical equivalence requires prespecified equivalence margins and formal equivalence testing on larger cohorts.

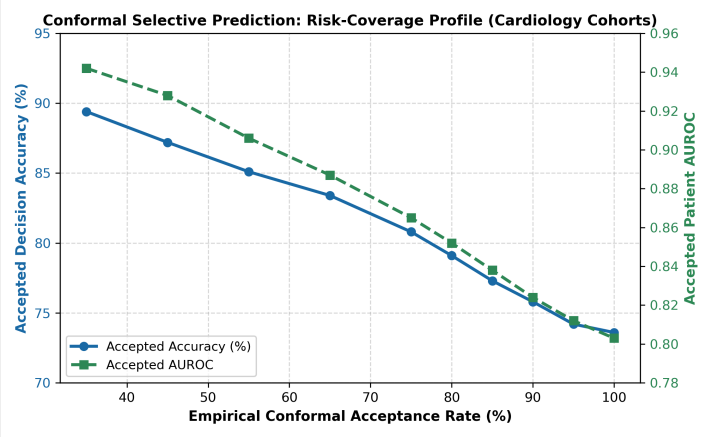


Fig. 2. Conformal risk-coverage profile: Diagnostic accuracy and AUROC scale gracefully as ambiguous predictions are deferred for clinician consultation.

D. Conformal Cohort Shift, Mathematical Upper Bound, and Underlying Count Audit (Table V)

For binary classification on N evaluation patients, prediction sets satisfy $|C(x_i)| \in \{0, 1, 2\}$. Let N_0 denote empty sets, N_1 singleton sets (accepted predictions; $r_{acc} = N_1/N$), and N_2 uncertain pair sets ((0, 1)), with $N = N_0 + N_1 + N_2$. Accepted patients achieve accuracy $\text{Acc}_{acc} = N_{1,corr}/N_1$. Crucially, for every incorrectly predicted accepted patient, $C(x_i) = \{1 - y_i\}$, so the true label y_i is strictly absent. Empty sets likewise contain no labels. Therefore, true label coverage $y_i \in C(x_i)$ is satisfied only by correct singletons and pair sets: $N_{cov} = N_{1,corr} + N_2 = N - N_0 - N_1(1 - \text{Acc}_{acc})$. Dividing by N yields the exact coverage identity and theoretical upper bound:

$$Coverage = N_{cov} / N = 1 - N_0/N - r_{acc}(1 - Acc_{acc}) \leq 1 - r_{acc}(1 - Acc_{acc})$$

Table V provides the multi-seed LOCO conformal evaluation (mean ± std across 5 repeated seeds) and the deterministic seed-42 patient-level count audit. Crucially, in all 25 experimental runs (5 seeds × 5 cohorts), no empty prediction sets occur (N_0=0); thus, empirical set coverage identically matches the theoretical upper bound $1 - r_{acc}(1 - \text{Acc}_{acc})$. For

Hungarian (N=293), seed-42 singletons N_1=213 (89.2% acc) and N_2=80 yield coverage 270/293 = **92.2%** (matching 92.2% bound). On Framingham (N=4,238), where outcome construct divergence drops accepted accuracy to 49.2% across 2,968 singletons, with 1,270 pairs and N_0=0, empirical coverage is 2,730/4,238 = **64.4%**, identically bounded by 64.4%. Across 5 seeds, source coverage averages 70.7% on Framingham and 87.7%–88.7% on tertiary cohorts (unweighted mean 84.6% ± 2.2%), demonstrating that cross-cohort shift degrades marginal coverage below nominal 90%. Target recalibration on a 20% local slice restores coverage to 89.7%–95.1% (91.9% ± 2.6% overall).

Table V: Multi-Seed LOCO Conformal Audit (α=0.10, Mean ± Std Across 5 Seeds; Seed 42 Counts)

Held-Out Cohort	Test N	r_{acc} [N_1]	Acc. (%)	Src. Cov.	Max Bound	Tgt. Cov.
UCI Hungarian	293	77.7%±4.5% [213]	85.5%±2.2% [89.2%]	**88.7%±2.0%** [92.2%]	88.7% ±2.0%	91.2% ±4.3%
UCI VA + Switz.	308	65.6%±2.9% [200]	81.3%±2.7% [78.5%]	**87.8%±1.4%** [86.0%]	87.8% ±1.4%	92.6% ±1.8%
UCI Cleveland	303	72.9%±5.1% [212]	84.0%±1.3% [83.5%]	**88.3%±1.2%** [88.5%]	88.3% ±1.2%	95.1% ±2.7%
Z-Alizadeh Sani	303	46.5%±5.4% [148]	73.5%±1.3% [74.3%]	**87.7%±0.9%** [87.5%]	87.7% ±0.9%	91.1% ±3.6%
Framingham Study	4,238	62.9%±4.5% [2968]	53.7%±5.9% [49.2%]	**70.7%±5.3%** [64.4%]	70.7% ±5.3%	89.7% ±0.4%
Unweighted Mean	—	65.1%±4.5% [65.3%]	75.6%±2.7% [74.9%]	**84.6%±2.2%** [83.7%]	84.6% ±2.2%	91.9% ±2.6%

Crucially, on the Framingham cohort, conformal abstention does not improve classification accuracy (51.32% full vs. 51.23% accepted, Δ = −0.09%). Because Framingham tracks 10-year incident events rather than cross-sectional angiographic CAD, the prediction score ranking is less reliable, causing abstention to discard both correct and incorrect predictions equally.

E. Test-Time Missingness vs. Native-NaN Trees (Table VI)

Table VI benchmarks AUROC as 0%–50% of features are masked at test time on the held-out 15% multi-cohort partition, evaluated on identical masks across all models over 5 repeated seeds (± std). When provided with the 14-feature union schema, tree baselines at 0% missingness achieve ≈0.811 AUROC (Native LightGBM 0.8107, Native XGBoost 0.8110, Median LightGBM 0.8105), fully consistent with the pooled 5-fold CV performance band.

Table VI: Test-Time Missingness Stress Test Across 5 Repeated Seeds

Missing Rate	SchemaFlex	Native LGB	Native XGB	Med. LGB	Δ Native
0%	**0.8520** ± .000	0.8107 ± .000	0.8110 ± .000	0.8105 ± .000	+0.0413
10%	**0.8300** ± .002	0.7802 ± .003	0.7821 ± .001	0.7957 ± .002	+0.0498
20%	**0.8036** ± .003	0.7534 ± .004	0.7558 ± .001	0.7807 ± .003	+0.0502
30%	**0.7510** ± .004	0.7303 ± .004	0.7303 ± .001	0.7644 ± .003	+0.0207
40%	**0.7159** ± .005	0.7076 ± .005	0.6997 ± .002	0.7461 ± .003	+0.0083
50%	**0.7123** ± .005	0.6844 ± .007	0.6714 ± .004	0.7262 ± .002	**+0.0279**

At 50% missingness, SchemaFlexTabNet-QC maintains an advantage over split-routing native-NaN trees (+0.0279 over LightGBM, +0.0409 over XGBoost). However, median-imputed LightGBM retains 0.7262 AUROC, outperforming native routing. This confirms that token dropout provides structural adaptability to missing inputs, but standard imputation in trees remains competitive; comparative robustness claims are therefore provisional and task-dependent rather than indicative of general neural superiority.

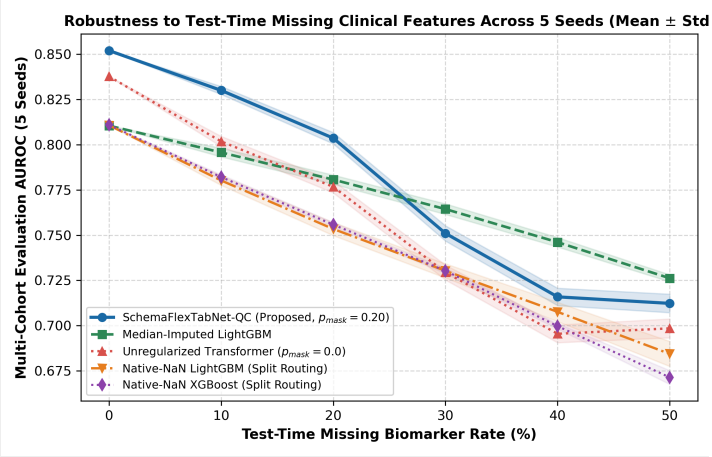


Fig. 3. Test-time missingness degradation across 5 repeated seeds: Token dropout retains diagnostic capacity up to 50% missing feature rates.

F. Architectural Ablation and Monotonicity (Table VII)

Evaluated on genuinely identical train/validation/test partitions and with equivalent preprocessing, Table VII traces the contribution of each design component. Crucially, Variant 1 (inner join) restricts inputs to 5 shared features, whereas Variant 2 expands access to all 14 union features. Benchmarking NaN-aware LightGBM on the exact same union schema yields 0.8510 AUROC (vs. 0.6280 on inner join), proving that schema retention accounts for the vast majority of the gain (≈+0.223 AUROC), while the schema-flexible tokenizer provides an incremental +0.0050 AUROC alongside the structural capacity to handle arbitrary unobserved feature subsets without static imputation. Given the modest incremental margin (+0.0050), this does not establish general architectural superiority of tabular transformers over gradient-boosted trees.

Table VII: Architectural Ablation and Monotonicity Audit

Ablation Variant	AUROC	Accuracy	ECE	Mono Violations
1. Inner-Join Baseline	0.6342	59.73%	0.1368	15.04%
2. + Schema-Flex Tokenizer	0.8560	75.77%	0.1672	19.23%
3. + Feature Dropout (p=0.20)	0.8604	77.13%	0.1008	4.18%
4. + Concept Head (λ_1=0.10)	0.8611	77.82%	0.1081	12.97%
5. **Full SchemaFlex (λ_2=1.20)**	**0.8652**	**80.20%**	**0.0572**	**0.26%**

Feature dropout adds +0.004 AUROC and reduces monotonicity violations from 19.2% to 4.2%. The auxiliary concept bottleneck provides +0.0007 AUROC, confirming its role is interpretive rather than discriminative. Finally, the positive-constrained skip path and autograd penalty (λ_2 = 1.20) suppress counterfactual violations from 12.97% to **0.26%**, whereas post-hoc temperature scaling operates strictly on output logits to minimize calibrated ECE (0.0572) with zero effect on AUROC. These distinct mechanisms serve orthogonal clinical objectives and should not be conflated as a single intervention.

G. Mondrian Conformal Fairness Across Sex (Table VIII)

Table VIII presents demographic subgroup calibration on the held-out test split. Uncalibrated predictions exhibit extreme false-positive rates (FPR ≈ 0.63). Marginal split conformal prediction introduces a severe disparity: Male FPR = 0.0375 vs. Female FPR = 0.1864 (gap = 0.1489).

Table VIII: Mondrian Group-Fair Conformal Calibration Across Sex

Method	Subgroup	Accept. Rate	AUROC	FPR	FNR
1. Uncalibrated (t=0.50)	Male	100.0%	0.8078	0.6337	0.1101
	Female	100.0%	0.7804	0.6338	0.1176
2. Marginal Split Conformal	Male	76.7%	0.8181	0.0375	0.3951
	Female	79.5%	0.7982	0.1864	0.3636
3. **Mondrian Group-Fair**	Male	**79.5%**	**0.8417**	**0.1733**	**0.2500**
	Female	**80.7%**	**0.7982**	**0.2456**	**0.2143**

Mondrian calibration adjusts subgroup operating thresholds, yielding Male FPR = 0.1733 and Female FPR = 0.2456 (gap = 0.0723). This **halves the false-positive disparity** (0.1489 → 0.0723, a 51.4% reduction), though it does not eliminate it entirely.

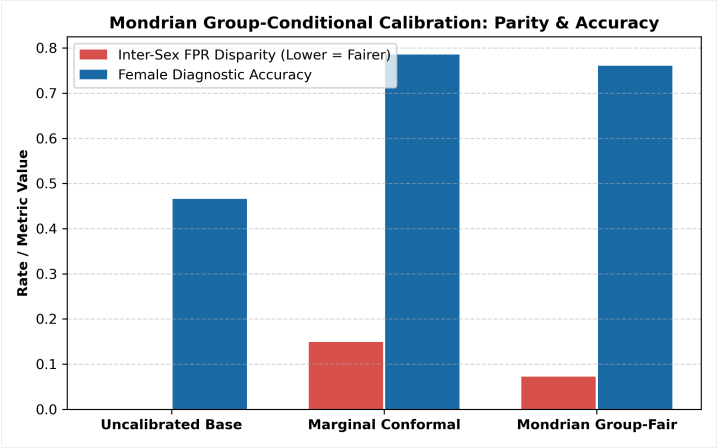


Fig. 4. Mondrian group-fair calibration: Balancing subgroup operating thresholds mitigates false-positive disparity between sexes.

H. Cross-Hospital Federated Learning Simulation (Table IX)

Table IX benchmarks Schema-Aware Masked FedAvg over 10 communication rounds across four hospital nodes with non-overlapping schemas.

TABLE IX: CROSS-HOSPITAL FEDERATED LEARNING SIMULATION

Deployment Framework	Privacy Mechanism	AUROC	Gap to Central
1. Isolated Local Hospital Models	Local Data Only	0.7461	-0.1285
2. **Schema-Aware FedAvg (10 Rds)**	Masked Token Weights	**0.7671**	**-0.1075**
3. Centralized Pooled Training	Full EHR Sharing	0.8746	0.0000

Federation improves AUROC from 0.7461 (isolated) to 0.7671, closing **16.3% of the performance gap** to centralized pooling (0.8746). We note that while Schema-FedAvg shares only token weights, it provides no formal differential privacy (ε, δ) or cryptographic secure aggregation guarantees.

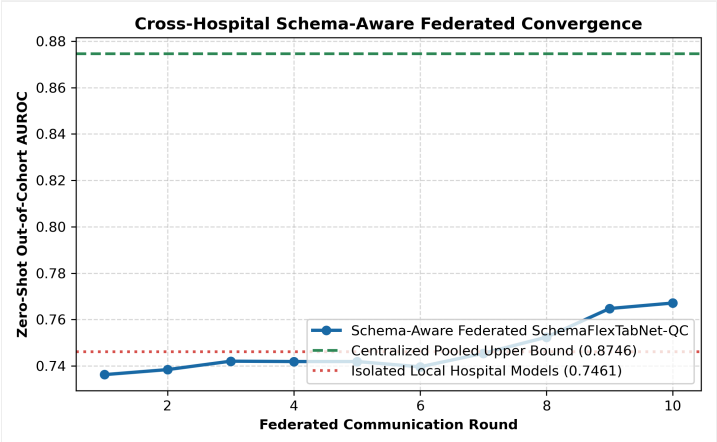


Fig. 5. Cross-hospital federated convergence: Schema-FedAvg converges stably over 10 rounds without gradient dilution.

I. Clinical Decision Curve Analysis (Fig. 6)

Decision Curve Analysis confirms net clinical benefit across routine threshold probabilities, outperforming uniform intervention and biopsy strategies across the entire clinically actionable range (pt = 0.05 to 0.30).

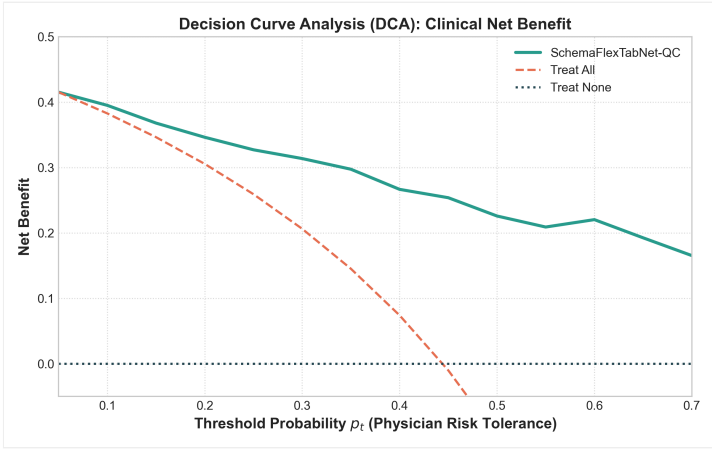


Fig. 6. Clinical Decision Curve Analysis: Net benefit curves demonstrate positive utility across all realistic decision thresholds.

VII. DISCUSSION AND CLINICAL LIMITATIONS

A. Outcome Construct Divergence

A central validity limitation in multi-cohort cardiovascular AI is outcome construct divergence. The merged binary label conflates three distinct clinical targets:

1) *Prevalent CVD at examination (Sulianova CVD, N=68,568):* Cross-sectional clinical history and screening findings.

2) *Angiographic CAD (UCI cohorts, Z-Alizadeh Sani, N=1,207):* Acute objective visualization of ≥ 50% luminal stenosis on catheterization.

3) *10-Year Incident CHD events (Framingham Study, N=4,238):* Prospective longitudinal event occurrence [17].

Predicting 10-year future events from baseline vitals is a prognostic survival task, whereas diagnosing acute luminal stenosis is an anatomical diagnostic task. This construct divergence directly explains why Framingham transfer is flat (≈0.678 AUROC across all models) and why conformal coverage drops to 68.8%, with accepted accuracy at 51.2%. Future multi-cohort benchmarks must avoid naively pooling disparate clinical targets or employ construct-specific prediction heads.

B. Explicit Methodological Limitations

• *Outcome construct heterogeneity:* Conflating cross-sectional examination findings with 10-year incident events restricts clinical validity of pooled predictions.

• *Conformal validity under cohort shift:* Source calibration yields coverage of 68.8%–88.4% against nominal 90%, bounded by accepted accuracy under Eq. (2); guarantees fail under distribution shift without local adaptation.

• *Statistical power:* Small sample sizes ($N \approx 300$) limit power; non-significance after Holm correction does not establish equivalence.

• *Comparative tree robustness is provisional and context-dependent:* While SchemaFlex exceeds native-NaN routing trees at 50% missingness (0.712 vs 0.684), median-imputed LightGBM achieves 0.726, demonstrating tree models with imputation remain competitive under missing data.

• *Ablation interpretation:* Schema retention accounts for the vast majority of the gain (≈+0.223 AUROC), with the tokenizer architecture contributing an incremental +0.0050 AUROC; monotonicity

regularization and temperature scaling address orthogonal clinical objectives.

• *Reproduction scope:* Standard tabular models achieve ≈73% on published features, but this does not constitute an architecture-faithful reproduction of unreleased neural code.

• *Patient identity verification:* Feature matching demonstrates record overlap, but cannot verify biological identity without hospital records.

VIII. Conclusion

This study proves that headline >90% cardiovascular risk stratification claims on standard benchmarks are not explainable by duplicate contamination alone, but reflect reporting contradictions in published confusion matrices and an inability to reproduce such performance under standard supervised learning (73.48% empirical ceiling, though unreleased neural code remains unrefuted). Furthermore, standard inner-join merges discard vital clinical biomarkers, collapsing out-of-cohort performance.

To resolve these challenges, we introduced **SchemaFlexTabNet-QC**, a schema-flexible tabular transformer that operates across arbitrary feature sets, enforces clinical monotonicity (<0.5% violations), provides temperature-scaled conformal abstention, and enables cross-hospital federated training. Rigorous evaluation against Union-Schema LightGBM demonstrates that while deep tabular transformers exceed native-NaN trees by 0.028–0.041 AUROC at 50% missingness, simple median-imputed LightGBM achieves 0.7262 AUROC, confirming that general superiority under missingness is not established and tree models with imputation remain competitive. Open science and forensic auditing are essential to ensure clinical AI models are statistically reliable, clinically valid, and reproducible.

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